Zheng Fang

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SUMMARY

Software engineer with a Harvard M.Des (Technology track), cross-registered at MIT EECS. I build AI-powered systems, backend services and full-stack web applications in Python, Java, Go and TypeScript, with hands-on work on RAG pipelines, vector search, computer vision and cloud deployment. Seeking a New Graduate Software Engineer role starting in 2026.

EDUCATION

Harvard University **Cambridge, MA**
M.Des (Technology track), Cross-registered at MIT EECS *Sep. 2024 - Jun. 2026*
Coursework: Fundamentals of Programming, Data Science I & II, ML Operations, Linear Algebra, Probability and Statistics, Data Visualization.

Syracuse University **Syracuse, NY**
Bachelor of Architecture - Magna cum laude *Sep. 2017 - Jun. 2022*

TECHNICAL SKILLS

Programming Languages: Java, JavaScript/TypeScript, Python, Go, SQL, C, C++, R
Databases and Cloud: PostgreSQL (Cloud SQL, pgvector), MySQL, Amazon RDS, AWS (Lambda, EC2, App Runner, Elastic Beanstalk), Google Cloud (GCE, GAE, GKE, Vertex AI), Docker, Kubernetes, GitHub Actions
Web Development: Spring Boot, Spring Security, Java Servlet, FastAPI, Flask, Node.js/Express, React, Redux, Vite, Mapbox GL JS, HTML & CSS, Linux
AI / ML & Data: LLM Integration (OpenAI API, Gemini), Retrieval-Augmented Generation (RAG), Vector Search (pgvector, FAISS), Recommendation Systems, Computer Vision (YOLO, CLIP, segmentation), Data Modeling & Feature Engineering

EXPERIENCE

Harvard Graduate School of Design **Cambridge, MA**
Research Associate, with Rachel Meltzer *May 2025 - Sep. 2025*

- Studied how street-level transparency relates to perceived safety across Boston; the work became the first-author CAADRIA 2026 paper *Prompts to Perception*.
- Built a **Python + GIS** pipeline that sampled street points citywide and collected **49,740 Google Street View images**.
- Applied **promptable segmentation** to extract street features (building facade and window-to-wall ratio, sky, trees, road, sidewalk) as model inputs.
- Collected pairwise safety ratings on 300 images, trained models to predict perceived safety for the whole city, and explained them with **SHAP**; sky, trees, land use, road, buildings and sidewalks carried the most weight.
- Ran spatial statistics (Moran's I of 0.3 at 500 m, LISA clusters) and produced citywide safety and window-to-wall ratio maps.

PROJECT EXPERIENCES

TopShot: An AI Photography Tutor with a RAG-Based Vision + Language Pipeline (team of 3)

- Developed an AI photography tutor that analyzes user-uploaded photos and delivers technical and composition feedback, powered by a **RAG-backed vision + language pipeline** over a curated dataset of about 25K photos.
- Built the backend with **FastAPI** and **PostgreSQL (Cloud SQL)**, covering **EXIF** metadata extraction, image embedding, and exemplar retrieval to ground feedback generation on **Vertex AI**.
- Wrote the retrieval layer: 768-d text embeddings stored in **pgvector**, top-k **cosine-distance** search on an **IVFFlat** index, with a similarity threshold to drop weak matches.
- Implemented the **TypeScript** frontend for photo upload and interactive feedback display.
- Containerized backend and frontend with **Docker Compose** and deployed to **Google Kubernetes Engine (GKE)** via Artifact Registry.
- Set up **CI/CD with GitHub Actions**: ruff linting, pytest unit/integration/e2e tests with a 60% coverage gate, image builds, and GKE deployment on every push to main.

TRACE: An Interactive Web Atlas of Acid Mine Drainage (React, Mapbox GL JS, Flask)

- Built all of the software, end to end, for an interactive map that links coal mining operations to acid mine drainage discharges, the stream segments they affect, and the water-quality stations that document the harm. Submitted to the **GSD Digital Design Prize 2026**.
- Built the frontend in **React**, **Vite** and **Mapbox GL JS**: browse and analyze modes, ranked search lists, pollution heatmaps, 3D extrusion columns with a 2D/3D switch, and entity panels with hand-drawn **SVG** sparklines for monitoring chemistry.
- Animated a downstream particle simulation in the browser with **requestAnimationFrame**, showing pollution spreading along a 20 km stream chain from up to 30 sources at once.
- Wrote an offline **Python** pipeline that merges PA DEP, USGS NHDPlus (20,033 stream segments), USGS NWIS and EPA Water Quality Portal data (about 75K chemistry samples), served through a **Flask** API with four blueprints and deployed on **AWS Elastic Beanstalk**.

OnlineOrder: A Spring Boot Based Online Food Ordering System

- Developed CRUD REST APIs with **Spring Controllers**, including registration, menu search, ordering, and checkout.
- Utilized **Spring Data JDBC** for **PostgreSQL** database access on **AWS RDS** (menus, restaurants).
- Implemented **Spring Security** for session-based authentication.
- Employed a structured architecture (controller, service, repository) with dependency injection.
- Built frontend using **ReactJS and Ant Design** for cart management and orders.
- **Containerized and deployed on AWS App Runner** for scalability.

LanternAI: Full-Stack RAG & MCP-Based Document Q&A Agent

- Designed and developed a full-stack AI agent using **React and Ant Design**, integrated with voice features via **react-speech-recognition** and **speak-tts** for spoken interaction.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** leveraging **GPT-5** and **LangChain** components for semantic document comprehension and context-aware responses.
- Designed a hybrid search strategy combining RAG outputs with web search through a **Model Context Protocol (MCP) server**, standardizing tool access and integrating results from **SerpAPI (Google Search API)**.
- Built and deployed a backend service with **Node.js and Express**, supporting document ingestion, query handling, and API orchestration.

MosaicAI: An AI-Based Social Network

- Designed and implemented a social network web application with **ReactJS**.
- Used OpenAI's **DALL·E 3** to assist users to create and update posts.
- Improved the authentication using token-based registration/login/logout flow with **React Router v4** and server-side user authentication with **JWT**.
- Launched a scalable web service in **Go** to handle posts and deployed to **Google Cloud (Google App Engine)**.
- Used **Elasticsearch** (deployed to GCE) to provide search functions such that users can search recent posts and list personal posts.

RESEARCH AND PUBLICATIONS

- Fang, Z., Li, X.
  Prompts to Perception: A Promptable Segmentation Approach to Transparency and Safety.
  Proceedings of CAADRIA 2026.
- Fang, Z.*, Tan, Y.*, Hoberman, C. (*equal contribution)
  TopoFold: Transformable Structure Generation from Topologically Complex Surfaces via Normal Intersection Search.
  Proceedings of CAADRIA 2026, Vol. 1, pp. 367–376.
- Fang, Z., Li, X., Sun, M., Tian, R.
  Sectional Sampling: Controlled Generation of High-Resolution Architecture Design with Diffusion Model and Latent Interpolation.
  Proceedings of ACADIA 2024, Vol. 2, pp. 623–632.
- Anonymous Authors
  Vinema: A Tangible, Vinyl-Inspired Device for Human–AI Musical Co-Creation.
  Manuscript under review, 2025.